SECTION 3 – SERVICE

3–1 INSTALLATION

Install Compact PDU using the procedures of this section. Complete installation in accordance with NEMA standards and local electrical code as appropriate.

3–1–1 Unpacking and Positioning

- The Compact PDU's are shipped on integral casters, wrapped in plastic. Go to Step NO TAG.

Note

If shock watch indicator is red, Compact PDU may have been damaged during shipment. Notify carrier immediately.

1. Remove cardboard cover.
2. Roll to installation site on integral casters.

CAUTION

Compact PDU weighs about 1000 lbs (455 kg). Use care when rolling Compact PDU. It weighs 250 lbs (114 kg) per caster. Plywood or steel plates may be used to distribute weight and protect floor surface from overload damage.

3. Remove protective plastic film.
4. Before final positioning of Compact PDU, verify that all cable routing conduit and raceway is positioned properly. If Compact PDU final position does not provide access to back or a side, route installation cables into cabinet before final positioning.

Note

For installation, 3 ft. (91 cm) of unobstructed space is needed for access to one side or back of Compact PDU. Access through front door is only access needed for periodic maintenance and repair.

Note

Models 46–317099P12: Taps on transformer are provided to permit setting the range of the units input voltage to match facility line voltage. Refer to tap diagram shown in illustration 3–14 for proper connection. For a mobile installation, taps are brought to an external switch near shore power connection. For a fixed–site installation the position of the transformer taps may need to be changed. In this case the left side of Compact PDU needs to be accessed for tap installation.

3–1–1 Unpacking and Positioning (continued)

5. Move Compact PDU to its final position.

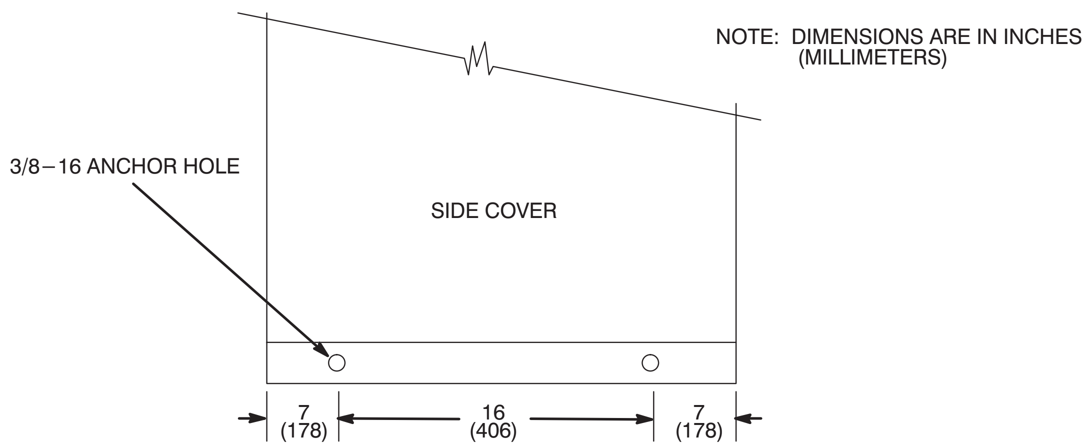
6. For a mobile installation, or if anchoring is required by local seismic code:

   a. Secure to floor using floor anchors.

      – Mark floor anchor prior to drilling. Hold floor anchor securely to base of Compact PDU. Using transfer bolt in documentation container inside outer door, inside Compact PDU strike transfer bolt through each Compact PDU anchor hole to mark floor anchor. See Illustration 3–1 for position of anchor holes in Compact PDU.

      – Drill anchor hole in floor anchor at each mark.

      – Bolt floor anchor to Compact PDU and to floor.



LOCATE FLOOR ANCHORS
ILLUSTRATION 3–1

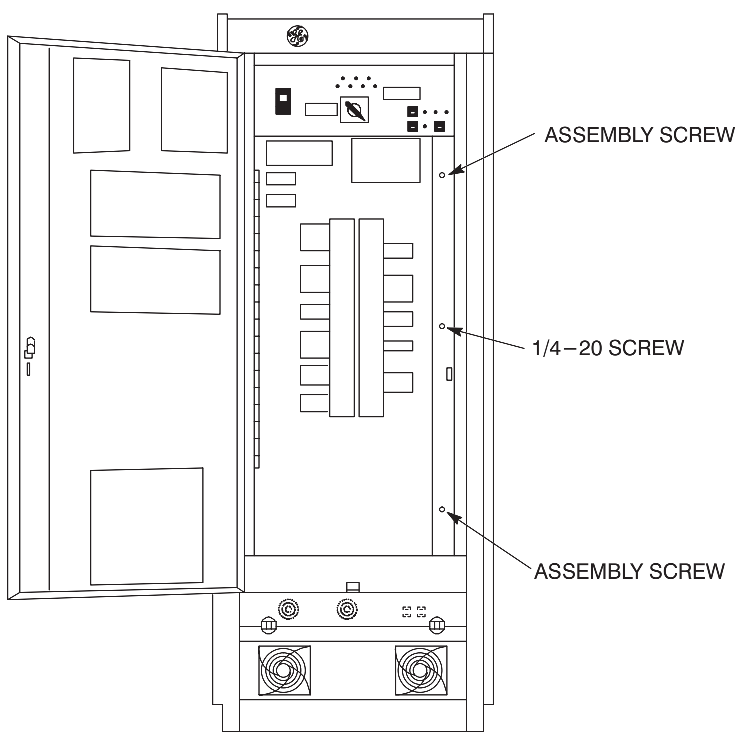
   b. Secure top to wall using wall anchor. Remove 2 eye bolts on top of Compact PDU nearest wall. Bolt angle bracket using 2 tapped holes which held eye bolts.

3–1–2 Facility Power and Dedicated Ground Connection

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! TO PREVENT FATAL ELECTRIC SHOCK, DISCONNECT POWER FROM COMPACT PDU AND LOCK OFF BEFORE YOU PERFORM THE FOLLOWING PROCEDURE.

1. Turn facility circuit breaker OFF, lock and tag.
2. Open outer door of Compact PDU.
3. Measure input voltage test points for 0 VAC before proceeding – if not zero – recheck step 1.
4. Loosen load panel door with a 90° turn of two assembly screws on right. See Illustration 3–2.
   - **Fixed site installation.** Remove and discard 1/4–20 screw in center right edge of door.
   - **Mobile installation.** Remove and keep 1/4–20 screw in center right edge of door.



OPEN LOAD PANEL DOOR
ILLUSTRATION 3–2

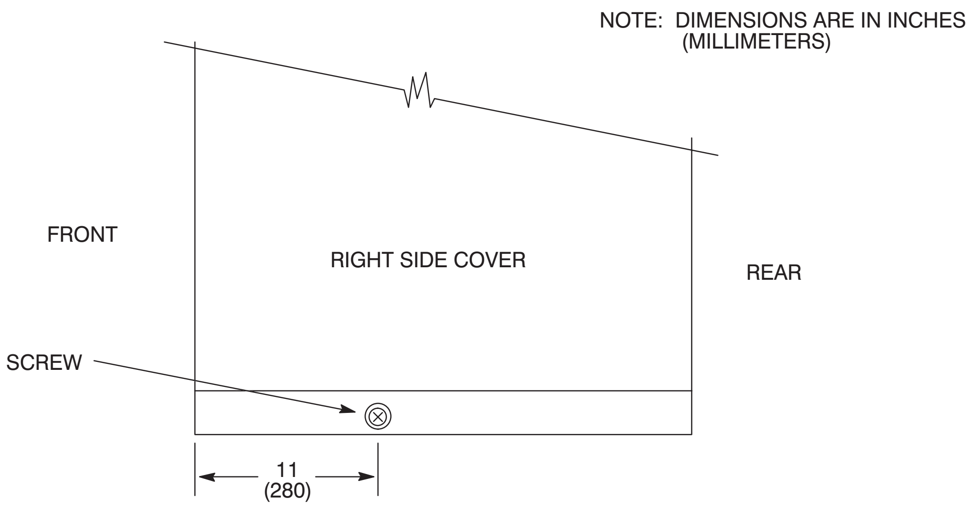
3–1–2 Facility Power and Dedicated Ground Connection (continued)

5. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.

6. Remove a side or back panel as follows:

   a. For side panel, remove screw under lower lip. See Illustration 3–3. Grasp both sides of panel and lift up firmly to remove.

   b. For rear panel, remove three screws on each side of panel. Support panel while removing seventh screw, located top center.



REMOVE SIDE PANEL
ILLUSTRATION 3–3

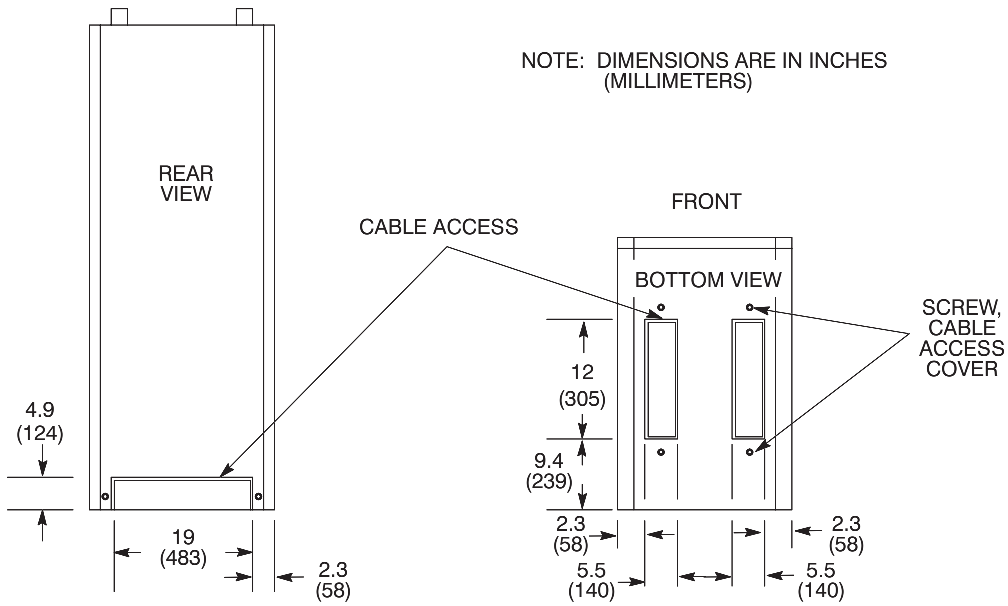
7. Remove a cable access cover. See Illustration 3–4 for cable access location.

   a. Cable access from bottom: Loosen screw on each end of cable access cover. Slide cable access cover to right, then swing up.

   b. Cable access from back: Remove screw on each side of cable access cover. Slide cable access cover down so it clears back cover, and remove.

Cable access cover may be discarded after a permanent installation, or it may be punched or drilled for appropriate conduit fittings.

3−1−2 Facility Power and Dedicated Ground Connection (continued)



CABLE ACCESS
ILLUSTRATION 3−4

8. Route facility input power and ground cables from conduit or cable opening into Compact PDU. Route to terminal block 10 (TB 10) (A2) and ground bus bar. See Illustration 3−5. Install four conductors to input terminal block on TB 10 (A2) (phase A, B, C, and neutral).

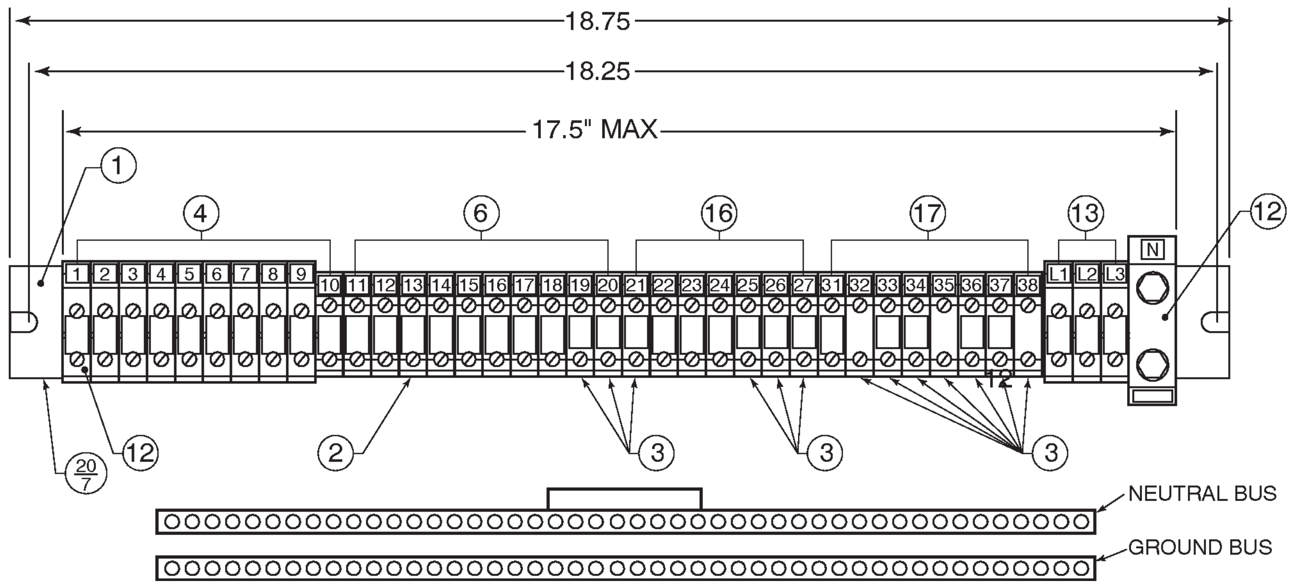
Note
Route all wires away from transformer. Use extra care if routing wires through base of Compact PDU.

9. Connect facility dedicated ground conductor (green/yellow) to 3/8−16 screw connection attached to ground bus bar on inside back wall of Compact PDU, below TB 10 (A2). See Illustration 3−5.

Note
Verify that dedicated ground is same size as incoming power conductors (at least AWG #1/0), or as required by local code.

Note
The neutral on TB 10 (A2) is the facility neutral hookup. If facility does not have a neutral LEAVE THIS CONNECTION OPEN. See Illustration 3−5.

3–1–2 Facility Power and Dedicated Ground Connection (continued)



TB 10 (A2) AND BUS BARS
ILLUSTRATION 3–5

10. Inspect all power connections and torque if needed. Refer to Section 3–6–4, PM Procedure, Steps 10 and 16.
11. Check connectors on circuit boards to verify proper seating. Refer to Section 3–6–4, PM Procedure, Step 17.
12. Refer to *Direction 15400, Signa Advantage System*, Set Up and Calibration tab, and perform Section 3, LINE VOLTAGE CHECKS.
13. Verify that incoming power meets requirements. Refer to *Direction 15402, Signa Advantage 1.5T, 1.0T, & 0.5T Site Planning*, Section 5, POWER REQUIREMENTS.
14. Models 46–317099P12 only: Move factory–installed transformer jumpers if needed.

Note
Use the factory–installed jumpers.
Only move factory–installed jumpers if changing from factory–set input voltage to another input voltage.
Refer to Section 3–2, CONFIGURATION, to determine which configuration is needed.

15. Turn all load panel circuit breakers OFF.

Note
Route all wires away from transformer. Use extra care if routing wires through base of Compact PDU.

16. Loosen cable clamp. See Illustration 3–5. Routing output wires away from transformer, bring them into Compact PDU through rear or a bottom cable access. Thread output wires through cable clamp to TB 10 (A2).

3−1−2 Facility Power and Dedicated Ground Connection (continued)

17. Connect output cables to terminals of TB10 (A2). Refer to Illustration 3−6, 3−7, 3−8, and 3−9.
18. Connect each cable safety ground wire to ground bus, below TB10 (A2), and neutral wire to neutral bus. See Illustration 3−5.
19. Connect equipotential stud on back of Compact PDU to next cabinet.

CAUTION

After all connections are made and verified correct, tighten each lug securely. This should be checked a second time, preferably by another person. Be sure all lugs are properly connected and securely tightened. High voltage does not forgive wiring errors or loose connections!

20. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4−20 screw in center right edge of door.
21. Unlock facility circuit breaker and turn ON.
22. Turn main circuit breaker ON.
23. Press Emer. Stop Reset switch.
24. Turn load panel circuit breakers ON.
25. Close outer door.

3–1–2 Facility Power and Dedicated Ground Connection (continued)

CB NO.	LOAD UNITS	TB10 UNIT	AMPERAGE		AMPERAGE	TB10 UNIT	LOAD UNITS	CB NO.
CB6	GRADIENT AMPLIFIER CABINET (208Y/120)		SHUNT		NONE	20	120 VAC	CB3 (SPARE)
		1	60A		NONE	21	120 VAC	CB4 (SPARE)
		2	60A		30A	22	INDEPENDENT CONSOLE (208Y/120)	CB5
		3	60A		30A	23		
CB7 (OPTION)	BROADBAND RF POWER AMPLIFIER		SHUNT		30A	24		
		4	NONE		SHUNT	99	SPARE	CB13
		5	NONE		NONE	7		
		6	NONE		NONE	8		
CB16	SUPER CON SHIM POWER SUPPLY (208Y/120)	33	30A		NONE	9		
		34	30A		30A	36	MAGNET POWER SUPPLY (208Y/120)	CB15
		35	30A		30A	37		
CB17	RF AMPLIFIER CABINET (208Y/120)		SHUNT		30A	38		
		10	30A		SHUNT		PENETRATION CABINET	CB8
		11	30A		40A	31		
		12	30A		15A	32	S/OUTL (120 VAC)	CB9
CB1	OPERATOR CONSOLE (208Y/120)	13	30A		SHUNT		RESISTIVE SHIM POWER SUPPLY (OPTION) (208 VAC)	CB18
		14	30A		15A	26		
		15	30A		15A	27		
CB11	SYSTEM/ COMPUTER CABINETS (208Y/120)	16	30A		15A	19	MFC (120 VAC)	CB2
		17	30A		15A	N/A	PDU	CB14
		18	30A		NONE	25	SPARE	CB12 (OPTION)

Shaded area numbers represent TB10 terminal numbers

LOAD CENTER SILK SCREEN (MODEL 46–317099P8/P9)
ILLUSTRATION 3–6

3−1−2 Facility Power and Dedicated Ground Connection (continued)

<table>
<tr><th>CB NO.</th><th>LOAD UNITS</th><th>TB10 UNIT</th><th>AMPERAGE</th><th></th><th>AMPERAGE</th><th>TB10 UNIT</th><th>LOAD UNITS</th><th>CB NO.</th></tr>
<tr><td rowspan="4">CB6</td><td rowspan="4">GRADIENT AMPLIFIER CABINET #1 208Y/120</td><td></td><td>SHUNT</td><td></td><td>15A</td><td>20</td><td>HEAT EXCHANGER</td><td>CB3</td></tr>
<tr><td>1</td><td>60A</td><td></td><td>NONE</td><td>21</td><td>SPARE</td><td>CB4</td></tr>
<tr><td>2</td><td>60A</td><td></td><td></td><td>22</td><td rowspan="3"></td><td rowspan="3">CB5</td></tr>
<tr><td>3</td><td>60A</td><td></td><td></td><td>23</td></tr>
<tr><td rowspan="4">CB7</td><td rowspan="4">GRADIENT AMPLIFIER CABINET #2 208Y/120</td><td></td><td>SHUNT</td><td></td><td></td><td>24</td></tr>
<tr><td>4</td><td>60A</td><td></td><td>30A</td><td>7</td><td rowspan="4">GRAM CABINET 208Y/120</td><td rowspan="4">CB13</td></tr>
<tr><td>5</td><td>60A</td><td></td><td>30A</td><td>8</td></tr>
<tr><td>6</td><td>60A</td><td></td><td>30A</td><td>9</td></tr>
<tr><td rowspan="3">CB16</td><td rowspan="3">SUPER CON SHIM POWER SUPPLY 208Y/120</td><td>33</td><td>30A</td><td></td><td>SHUNT</td><td></td></tr>
<tr><td>34</td><td>30A</td><td></td><td>30A</td><td>36</td><td rowspan="3">MAGNET POWER SUPPLY 208Y/120</td><td rowspan="3">CB15</td></tr>
<tr><td>35</td><td>30A</td><td></td><td>30A</td><td>37</td></tr>
<tr><td rowspan="4">CB17</td><td rowspan="4">RF AMPLIFIER/ CABINET 208Y/120</td><td></td><td>SHUNT</td><td></td><td>30A</td><td>38</td></tr>
<tr><td>10</td><td>30A</td><td></td><td>NONE</td><td></td><td rowspan="2">SPARE (120 VAC)</td><td rowspan="2">CB8</td></tr>
<tr><td>11</td><td>30A</td><td></td><td></td><td>31</td></tr>
<tr><td>12</td><td>30A</td><td></td><td>15A</td><td>32</td><td>S/OUTL (120 VAC)</td><td>CB9</td></tr>
<tr><td rowspan="3">CB1</td><td rowspan="3">OPERATOR CONSOLE 208Y/120</td><td>13</td><td>30A</td><td></td><td>NONE</td><td>26</td><td rowspan="3">RESISTIVE SHIM POWER SUPPLY(OPTION) (208 VAC)</td><td rowspan="3">CB18</td></tr>
<tr><td>14</td><td>30A</td><td></td><td>NONE</td><td>27</td></tr>
<tr><td>15</td><td>30A</td><td></td><td>NONE</td><td></td></tr>
<tr><td rowspan="3">CB11</td><td rowspan="3">SYSTEM/ COMPUTER CABINETS 208Y/120</td><td>16</td><td>30A</td><td></td><td>15A</td><td>19</td><td>MFC (120 VAC)</td><td>CB2</td></tr>
<tr><td>17</td><td>30A</td><td></td><td>15A</td><td>N/A</td><td>PDU (120 VAC)</td><td>CB14</td></tr>
<tr><td>18</td><td>30A</td><td></td><td>NONE</td><td>25</td><td>SPARE (120 VAC)</td><td>CB12</td></tr>
</table>

Shaded area numbers represent TB10 terminal numbers

LOAD CENTER SILK SCREEN (MODEL 46−317099P10/P11/P12)
ILLUSTRATION 3−7

3–1–2 Facility Power and Dedicated Ground Connection (continued)

CB NO.	LOAD UNITS	TB10 UNIT	AMPERAGE		AMPERAGE	TB10 UNIT	LOAD UNITS	CB NO.
CB6	GRADIENT AMPLIFIER CABINET #1 (208Y/120)		SHUNT		NONE	20	SPARE (120 VAC)	CB3
		1	60A		NONE	21	SPARE (120 VAC)	CB4
		2	60A		30A	22		CB5
		3	60A		30A	23		
CB7	MRT CABINET (208Y/120)		SHUNT		30A	24		
		4	30A		30A	9	MAGNET CABINET (208Y/120)	CB13
		5	30A		30A	7		
		6	30A		30A	8		
CB16	SUPER CON SHIM POWER SUPPLY (208Y/120)	33	30A		SHUNT			
		34	30A		30A	36	MAGNET POWER SUPPLY (208Y/120)	CB15
		35	30A		30A	37		
CB17	RF AMPLIFIER CABINET (208Y/120)		SHUNT		30A	38		
		10	30A		40A	31	PENETRATION CABINET (120 VAC)	CB8
		11	30A		SHUNT			
		12	30A		15A	32	S/OUTL (120 VAC)	CB9
CB1	OPERATOR CONSOLE (208Y/120)	13	30A		NONE	26	RESISTIVE SHIM POWER SUPPLY (OPTION) (208 VAC)	CB18
		14	30A		NONE	27		
		15	30A		NONE			
CB11	SYSTEM/ COMPUTER CABINETS (208Y/120)	16	30A		15A	19	MFC (120 VAC)	CB2
		17	30A		15A	N/A	PDU PWR (120 VAC)	CB14
		18	30A		NONE	25	SPARE (120 VAC)	CB12

Shaded area numbers represent TB10 terminal numbers

LOAD CENTER SILK SCREEN (MODEL 46–317099P13/P15)
ILLUSTRATION 3–8

3–1–2 Facility Power and Dedicated Ground Connection (continued)

CB NO.	LOAD UNITS	TB10 UNIT	AMPERAGE		AMPERAGE	TB10 UNIT	LOAD UNITS	CB NO.
CB6	GRADIENT AMPLIFIER CABINET #2 (208Y/120)		SHUNT		20A	20	208V OUTLET #1	CB3/CB4
		1	60A		20A	21		
		2	60A		NONE		SPARE	CB5
		3	60A		NONE			
CB7	SPARE		NONE		NONE			
			NONE		NONE		SPARE	CB13
			NONE		NONE			
			NONE		NONE			
CB16	SPARE		NONE		NONE			
			NONE		NONE		SPARE	CB15
			NONE		NONE			
CB17	SPARE		NONE		NONE			
			NONE		20A	31	120 OUTLET #7 (120 VAC)	CB8
			NONE		NONE		SPARE	CB9
CB1	120V OUTLET #1 (120 VAC)	13	20A		15A	26	208V OUTLET #2	CB18
	120V OUTLET #2 (120 VAC)	14	20A		15A	27		
	120V OUTLET #3 (120 VAC)	15	20A		SHUNT			
CB11	120V OUTLET #4 (120 VAC)	16	20A		20A	19	120V OUTLET #8	CB2
	120V OUTLET #5 (120 VAC)	17	20A		15A	N/A	PDU POWER	CB14
	120V OUTLET #6 (120 VAC)	18	20A		20A	25	120V OUTLET #9	CB12

Shaded area numbers represent TB10 terminal numbers

LOAD CENTER SILK SCREEN (MODEL 46–317099P14/P16)
ILLUSTRATION 3–9

3–1–2 Facility Power and Dedicated Ground Connection (continued)

3–1–3 Re Functional Checks

1. Open outer door of Compact PDU.
2. Turn main circuit breaker OFF.
3. Turn all load panel circuit breakers OFF.
4. Turn facility circuit breaker ON.
5. Press Emer. Stop Reset Switch.
6. Turn Main Circuit Breaker ON.
7. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center right edge of door. See Illustration 3–2. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.

3–1–3 Functional Checks (continued)

DANGER!!

FATAL SHOCK HAZARD!! USE EXTREME CAUTION FOR STEPS 8 THROUGH 18. LETHAL VOLTAGES EXIST ON LOAD CENTER BRACKET ASSEMBLY (A14), AND ON BOTH SIDES OF COMPACT PDU INTERIOR. ON MODELS 46–317099 P8 THROUGH P16 EXCEPT P12, LETHAL VOLTAGES EXIST ON THE REGULATION PANEL.

8. Check that voltage at input terminals of TB10 L1, L2, and L3 matches voltage on nameplate located over load panel.

9. Check that input current is phased in clockwise rotation (phase A, B, C). Use a phase meter. Refer to *Direction 15401, Signa Advantage Subsystems*, PDU: Functional Checks tab, and perform Section 1–1, PHASE SEQUENCE CHECK. Refer to Table 3–1 for phase relationship of input and output voltages.

TABLE 3–1
PHASE SEQUENCE

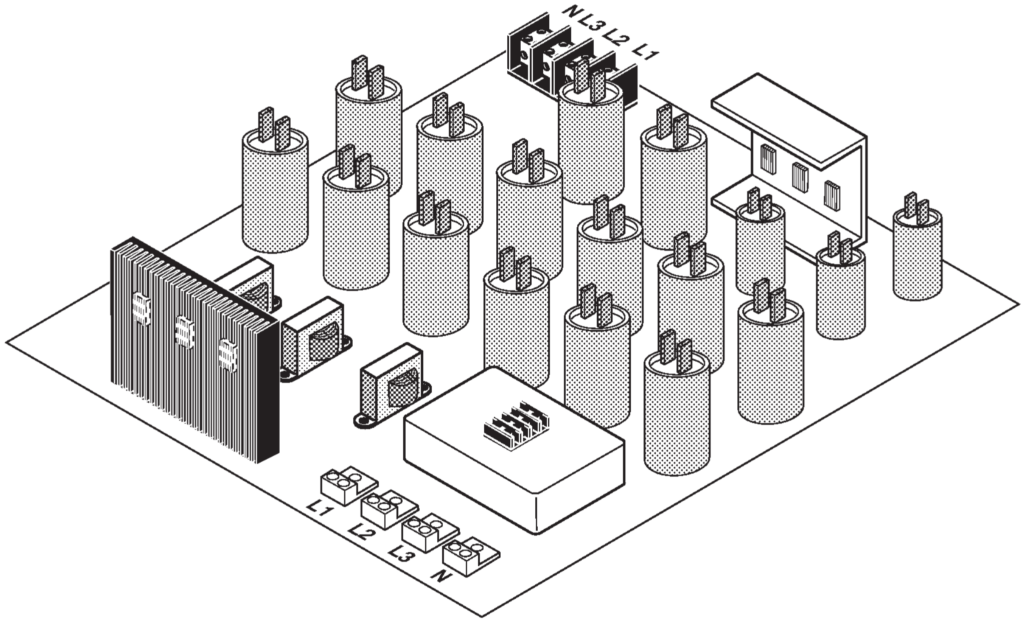
PHASE	1 (L1)	2 (L2)	3 (L3)
PHASE METER	A	B	C
WIRE COLOR	BROWN	RED	ORANGE
H	H_1	H_2	H_3
L	L_1	L_2	L_3
X	X_1	X_2	X_3

CAUTION

Voltage must be phased in clockwise rotation. Improper phase rotation can damage medical equipment.

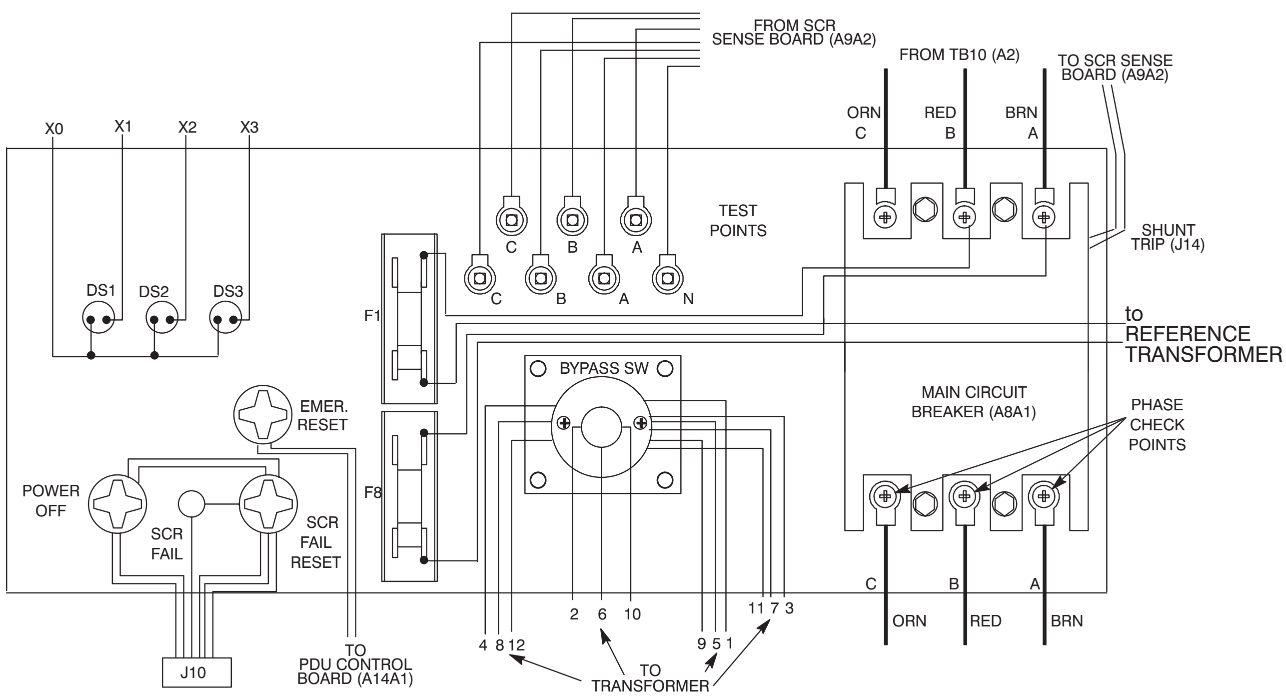
a. **Models 46–317099 P8 through P16 except P12.** Check phasing at primary terminals L1, L2, and L3 line side TB1 through TB4 of filter assembly. See Illustration 3–10.

3−1−3 Functional Checks (continued)



FILTER ASSEMBLY
ILLUSTRATION 3−10

b. **Models 46−317099 P8 through P16.** Check phasing at bottom of main circuit breaker. See Illustration 3−11.



FRONT PANEL, MODEL 46−317099 P8 THROUGH P16 EXCEPT P12 (REAR VIEW)
ILLUSTRATION 3−11

3–1–3 Functional Checks (continued)

10. Models 46–317099 P8 through P16 except P12 only: Measure facility line–to–line voltage at input test points and compare with model voltage specification given in Table 3–2.

Note

There is a 100:1 voltage stepdown at test points. For example, 480 VAC in Compact PDU is measured as 4.8 VAC at test points.

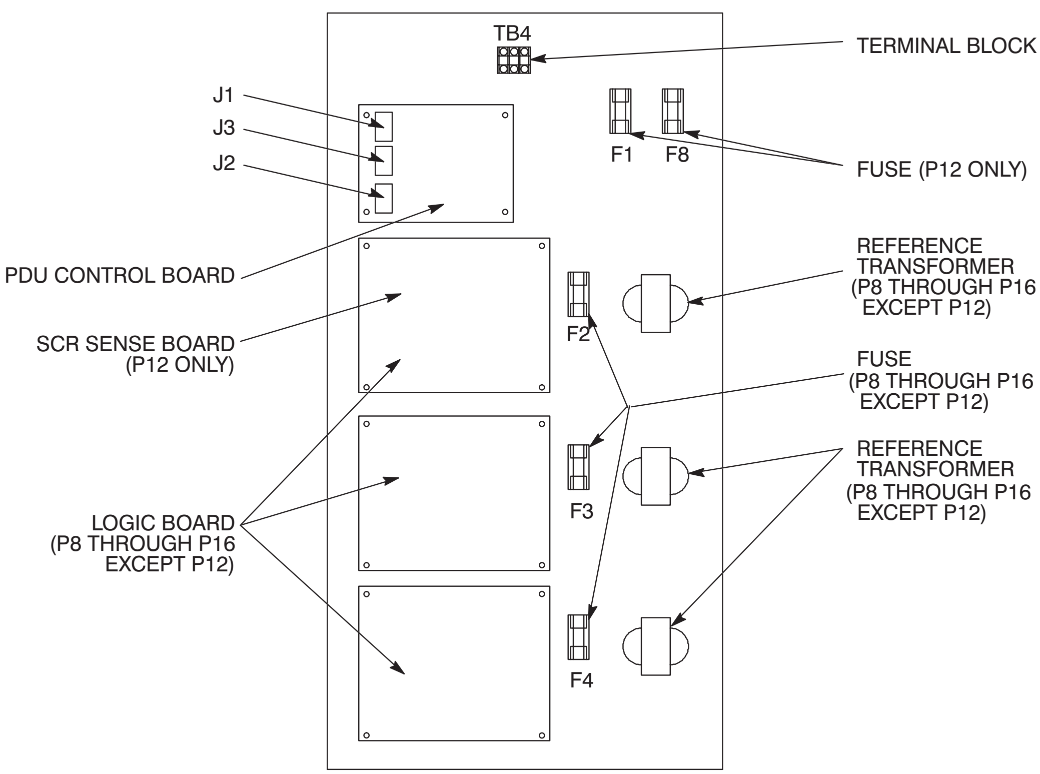
TABLE 3–2
VOLTAGE INPUT BY MODEL

COMPACT PDUPART NO.	NOMINAL INPUTLINE–TO–LINE	INPUT VOLTAGE AT INPUT TEST POINTS	INPUT VOLTAGE AT TB10, L1,L2,L3
46–317099P8/P9/P10/P15/P16	480 V 60 Hz regulated, filtered	4.18 to 5.04 V	412 to 502
46–317099P11/P13/P14	380, 400, or 415 V 50/60 Hz regulated, filtered	3.31 to 3.99 V, or 3.48 to 4.20 3.61 to 4.32	331 to 399 348 to 420 361 to 432

Note

Use the next step to perform functional checks without cables connected to J1, J2, and J3 of PDU control board.

11. In documentation package inside outer door, there is a jumpered connector. Insert jumpered connector into mating connector A14A1J2 on PDU control board. See Illustration 3–12.



LOAD CENTER BRACKET ASSEMBLY (A14)
ILLUSTRATION 3–12

3–1–3 Functional Checks (continued)

12. Turn main circuit breaker ON. After 6 seconds, Output indicator lights DS1, DS2 and DS3 light.
13. Press emer. stop reset switch. If indicator lights do not light, refer to Section 3–3, TROUBLESHOOTING.
14. Verify voltage of 208 V 3 phase with neutral at magnet and superconducting shim service outlets. See Illustration 1–4.
15. Verify that the two fans are running. Hold a piece of paper in front of each fan in succession. If paper does not stick to fan guard, check fan. Refer to Section 3–6–4, PM Procedure, Step 19.
16. If voltage is within specification, system is ready to run. If not, refer to Section 3–1–4, Calibrate Logic Board.
17. Turn load panel circuit breakers ON.
18. Verify that voltages on TB10 (A2) match output voltages shown in appropriate Illustration 3–6, 3–7, 3–8, or 3–9.
19. Prepare unit for service:
    a. Install any side or rear panels removed.
    b. Release articulated arm. Close load panel door. Secure load panel door with a 90˚ turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
    c. Close outer door.

3–1–4 Calibrate Logic Board (Models 46–317099 P8 through P16 except P12)

The purpose of this procedure is to set the output of regulator to load.

The driver board has seven green LEDs. Two LEDs should be lighted at all times. LED 7 indicates power on board and is always on. LEDs 1–6 represent tap settings and only one should be on at any time.

If no LED or one LED is lighted, or if more than two LEDs are lighted, immediately turn main circuit breaker OFF. Refer to Section 3–3, TROUBLESHOOTING, for diagnosis and repair.

If only one LED (1–6) is lighted, perform the following calibration:

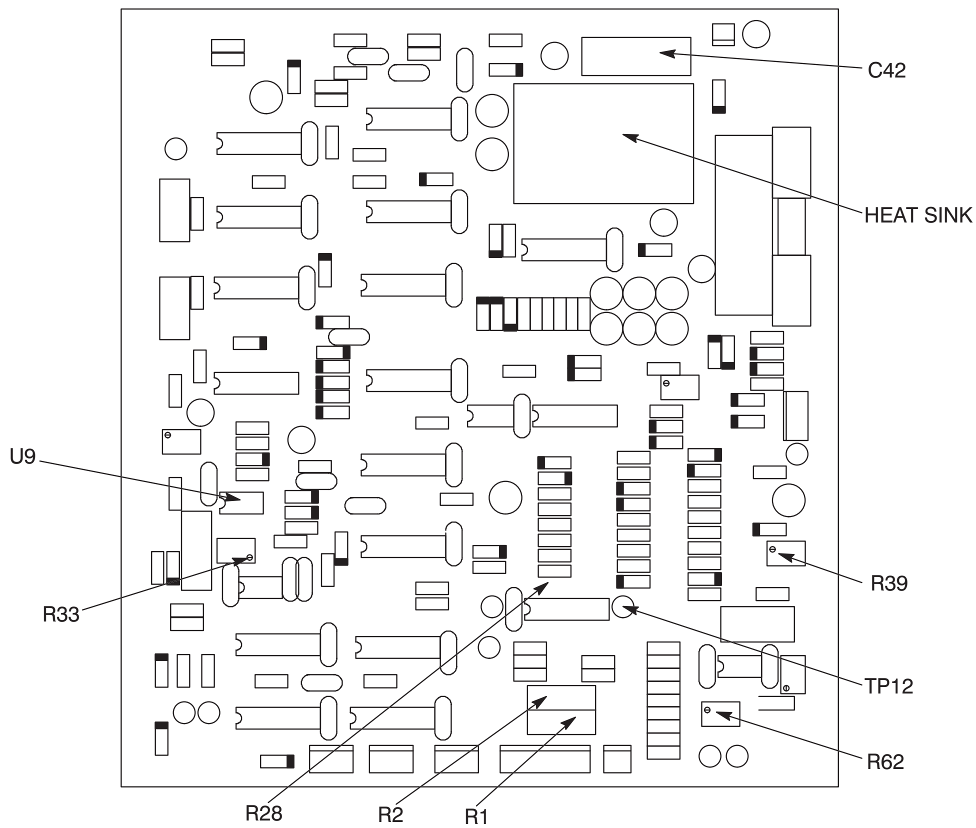
1. Adjust load.
    a. Turn on load and try to achieve 1 KW per phase (8.3 amps)

    Note

    Potentiometer R62 is a 30–turn pot. It may require several turns before output level changes.

    b. Verify voltage across C42 on logic board of 8.5–9.0 VDC. See Illustration 3–13.
    c. If reading is low, adjust R62 to increase reading. If correct reading cannot be obtained, refer to Section 3–4–2, Basic Check, Steps 11–13 to troubleshoot logic board.
    d. Perform step b. for phase B and C.

3−1−4 Calibrate Logic Board (Models 46−317099P8 through P16 except P12) (continued)



LOGIC BOARD
ILLUSTRATION 3−13

2. Adjust voltage.

   a. Locate R39 on phase A logic board.

   b. Verify output of phase A to neutral at test point of 1.20 $\pm$ 0.03 V. Measure with a true RMS multimeter.

   Note
   Output voltage does not change linearly as potentiometer R39 is rotated. Output voltage does a step change. R39 is a 30−turn pot. It may require several turns before output level changes.

   c. Adjust R39 to obtain correct output voltage for phase A.

   d. Perform Step 2 for phase B and phase C.

3−2 CONFIGURATION

3−2−1 Model 46−317099 P8 through P16 except P12

Model 46−317099 P8, P10, P15, and P16 are shipped factory configured. No configuration change is necessary. Do not change tap connections.

3−2−2 Model 46−317099P12

Model 46−317099P12 is shipped factory configured for 380 V nominal input. To change nominal input voltage:

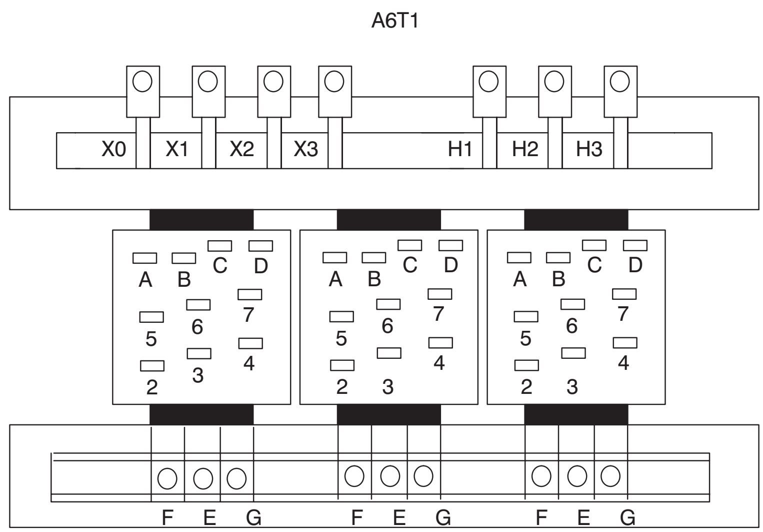
1. Connect the three transformer tap leads as shown in Illustration 3−14. Refer to Section 3−4−6, Change Transformer Tap−lead Position.

TAP CONNECTION

	NOMINAL INPUT	CONNECTION TAPS	
50 HZ	435 V	4 TO 5	AND B TO C
	425 V	4 TO 6	
	415 V	3 TO 5	
	405 V	3 TO 6	
	395 V	2 TO 6	
	385 V	3 TO 7	
	375 V	2 TO 7	
60 HZ	504 V	4 TO 5	AND B TO D
	492 V	4 TO 6	
	480 V	3 TO 5	
	468 V	3 TO 6	
	456 V	2 TO 6	
	444 V	3 TO 7	
	432 V	2 TO 7	
50 HZ	400 V	4 TO 5	AND A TO D
	390 V	4 TO 6	
	380 V	3 TO 5	
	370 V	3 TO 6	
	360 V	2 TO 6	
	350 V	3 TO 7	
	340 V	2 TO 7	

A, B, C, D TAPS FOR
480/415/400/380 V INPUT ONLY

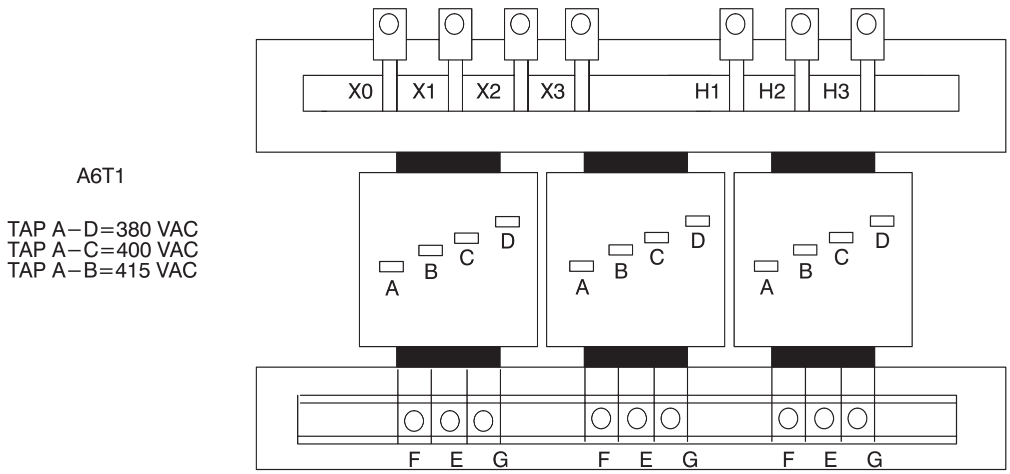
SECONDARY JUMPERS;
E − F for 60 HZ
E − G for 50 HZ



TRANSFORMER POWER CONNECTIONS, MODEL 46−317099P12
ILLUSTRATION 3−14

3−2−3 Model 46−317099P9, P11, P13, and P14

Model 46−317099P9, P11, P13, and P14 are shipped factory configured for 380/400/415 VAC, 50 HZ input.



TRANSFORMER POWER CONNECTIONS, MODEL 46−317099P9, P11, P13, AND P14
ILLUSTRATION 3−15